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INFORMATION SECURITY FINAL PROJECT REPORT

ANOMALY DETECTION IN NETWORK TRAFFIC USING MACHINE LEARNING

A COMPARATIVE STUDY OF ISOLATION FOREST AND OTHER MODELS

SUPERVISOR : HỒ HẢI

MEET OUR TEAM

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PROBLEM STATEMENT & OBJECTIVES

- **Problem:** Binary Classification (Benign vs. Malicious) on modern NetFlow data.
- **Models for Comparison:**
  - Baseline (Unsupervised): Isolation Forest (IF).
  - Supervised: Random Forest (RF), XGBoost (XGB).
- **Objectives:** Evaluate real-world performance via Fixed Thresholds, Top-K (Alert Budget), and Cross-dataset Generalization.

WHY THIS PROBLEM IS HARD IN PRACTICE

- Attack ≠ always clear anomalies
- Models can learn shortcuts (port, IP...)
- High accuracy ≠ real-world performance
- Domain shift breaks models

DATASET & GENERAL PIPELINE

- Dataset: NF-UNSW-NB15 v2 & v3
  - V2: 2,390,275 flows; 45 columns
  - V3: 2,365,424 flows; 55 columns
  - Malicious ratio:
    - V2: 3.9767%
    - V3: 5.3983%
  - Main task: binary classification (Benign vs. Malicious)

> **School of Information Technology and Electrical Engineering**
>
> The datasets on this page are designed for machine learning-based Network Intrusion Detection Systems (NIDS) and are organised into the following high-level collections
>
> School of Information Technology and Electrical Engineering

https://espace.library.uq.edu.au/view/UQ:6e0eda1

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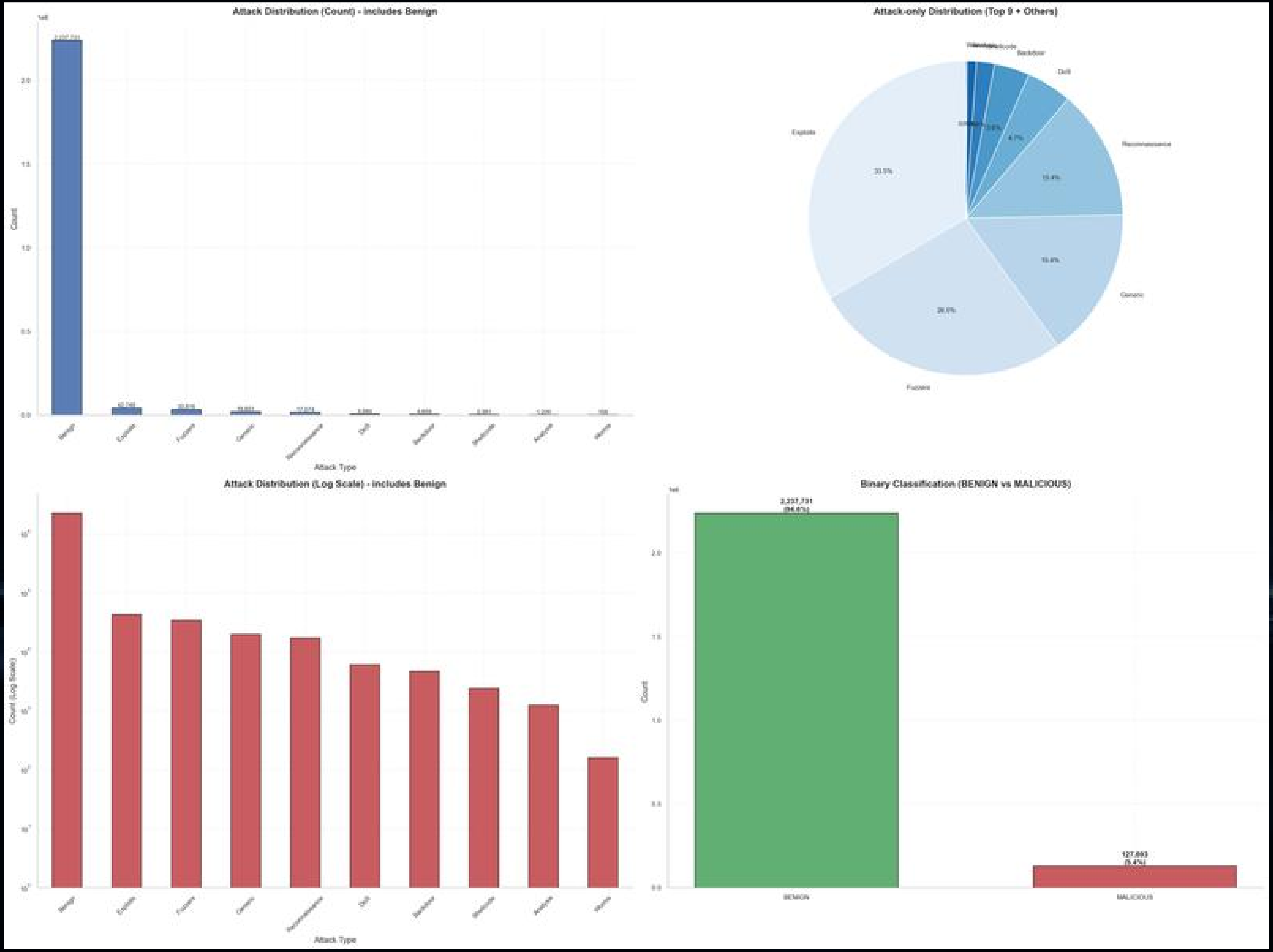
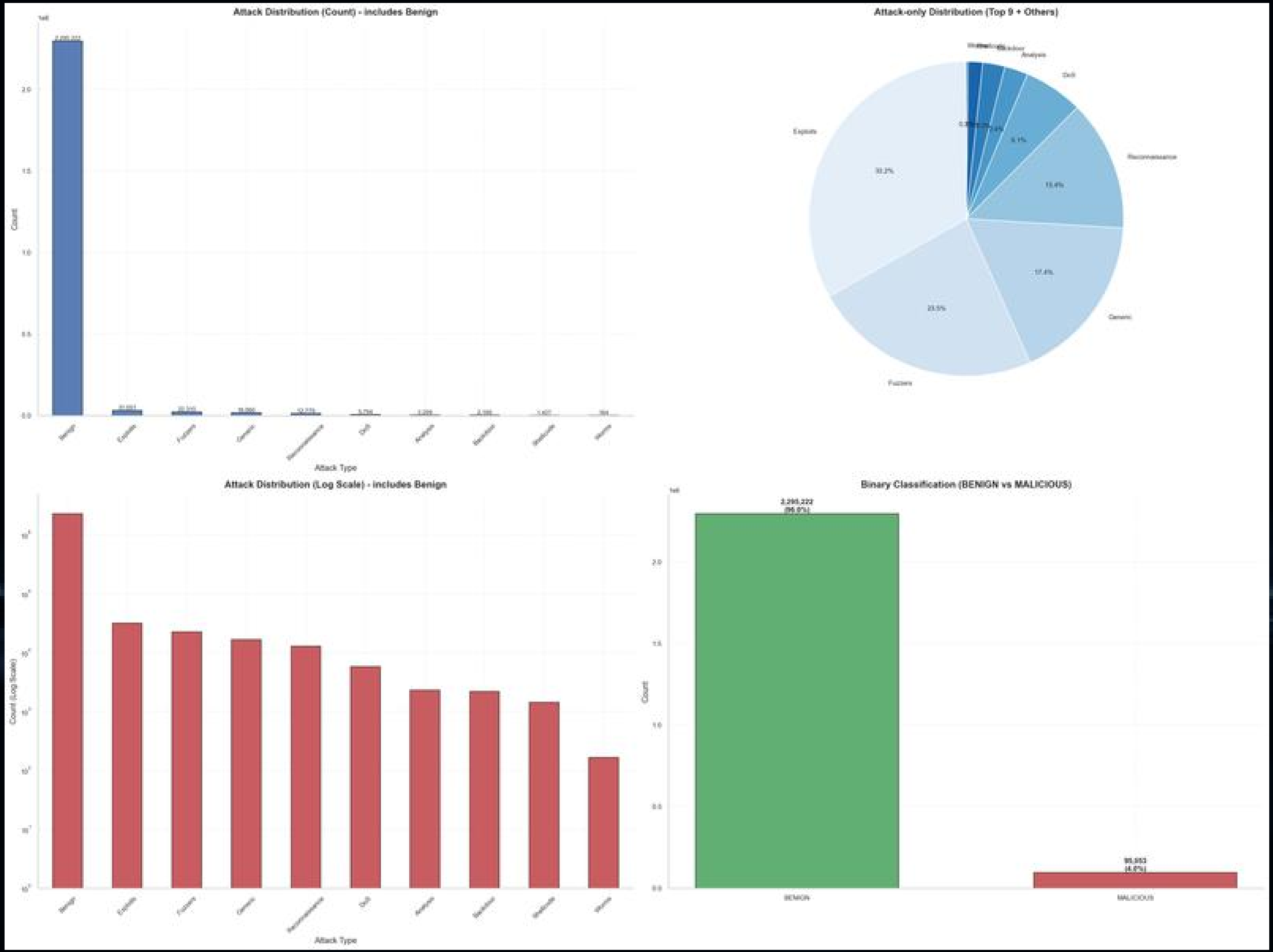
NF-UNSW-NB15 V3 ATTACK OVERVIEW



NF-UNSW-NB15 V2 ATTACK OVERVIEW



DATASET & GENERAL PIPELINE

- Dataset: NF-UNSW-NB15 v2 & v3

- Pipeline Stages:

1. Input: CSV --> Feature Engineering.
2. Canonical Split: Group split by IPV4_SRC_ADDR (Prevents leakage, ensures realistic evaluation).
3. Deduplication: Removing overlapping flows between Train and Test sets.
4. Evaluation: Performance metrics & visualization.

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CANONICAL SPLIT

NF-UNSW-NB15 V3

- Split by IPV4_SRC_ADDR
- Train groups: 32
- Test groups: 8
- No overlap between train and test groups
- After deduplication:
  - Train: 1,886,156 flows
  - Test: 413,167 flows
- Exact duplicate test rows removed: about 66,101

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WHY SPLIT BY SOURCE IP?

Avoid "Soft Leakage"

Prevents the same attacker/client behaviors from appearing in both sets.

Realism

Evaluates the model's ability to detect attacks from unseen IP sources.

Result

Reduces overfitting to identification features (shortcuts) and provides a more grounded accuracy score.

TREE-BASED MODELS: SAME STRUCTURE, DIFFERENT GOALS

Common Idea

- Binary splits on features
- Simple rules → partition data

Key Differences

- RF → many trees → stable average
- XGB → sequential trees → error correction
- IF → isolate points → detect anomalies

Same structure, different objectives → Different real-world behavior

ISOLATION FOREST (IF) - FROM THEORY TO WEAKNESSES

IF IN "EASY MODE" (NSL_KDD)

Why NSL_KDD?

Used as a baseline to demonstrate IF when anomalies are clearly separated from normal traffic.

Mechanism

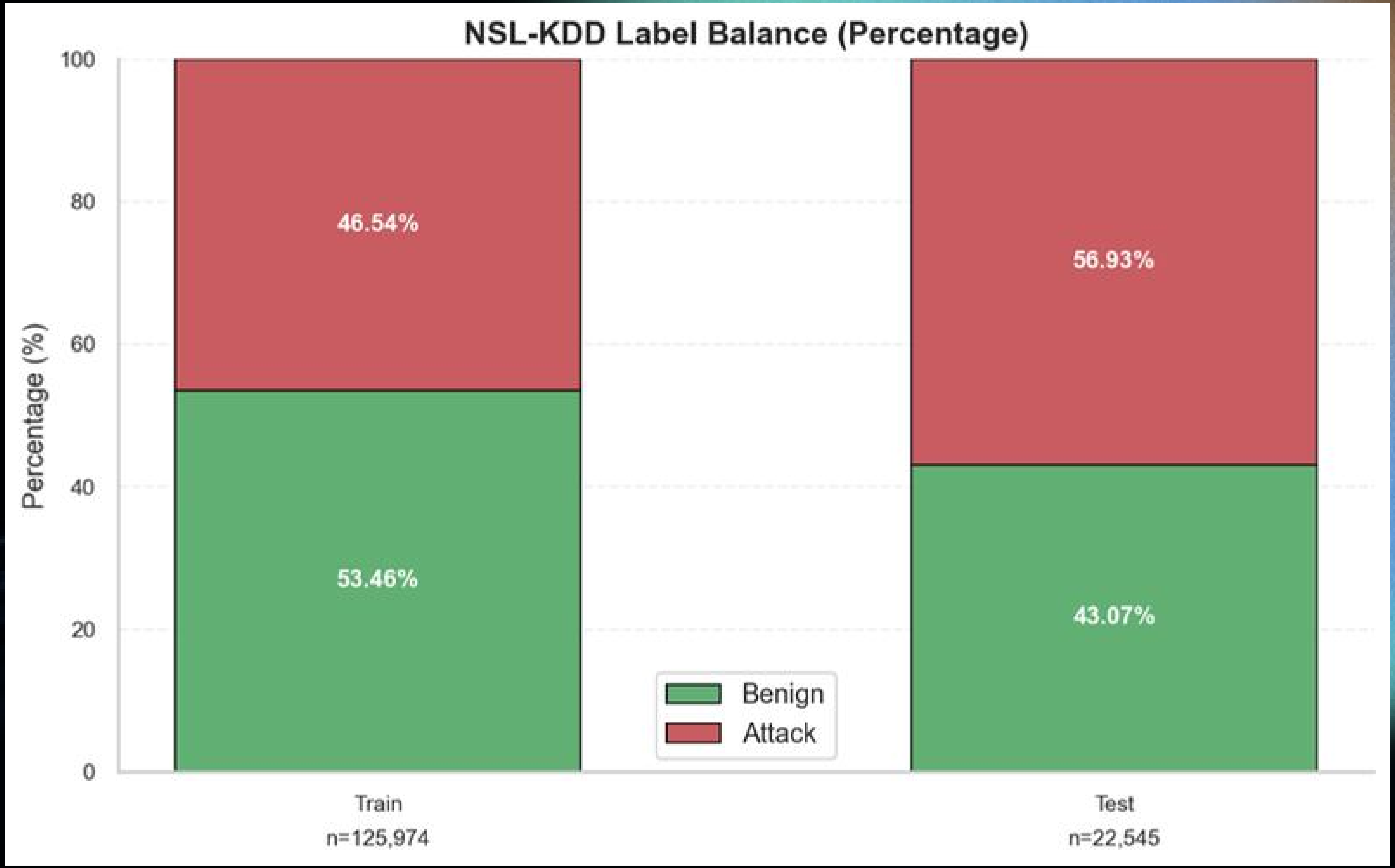
Anomalies (outliers) are easier to isolate --> shorter path in the tree --> more anomalous prediction.

IF MINI EXAMPLE

	salary	scores	anomaly
0	26100	0.121556	1
1	73188	0.063679	1
2	33333333333	-0.186302	-1
3	14623	0.144374	1
4	44444444444	-0.211409	-1
5	95759	0.113981	1
6	22368	0.147079	1
7	90715	0.020700	1
8	18507	0.119701	1
9	11559	0.104399	1
10	26800	0.146092	1

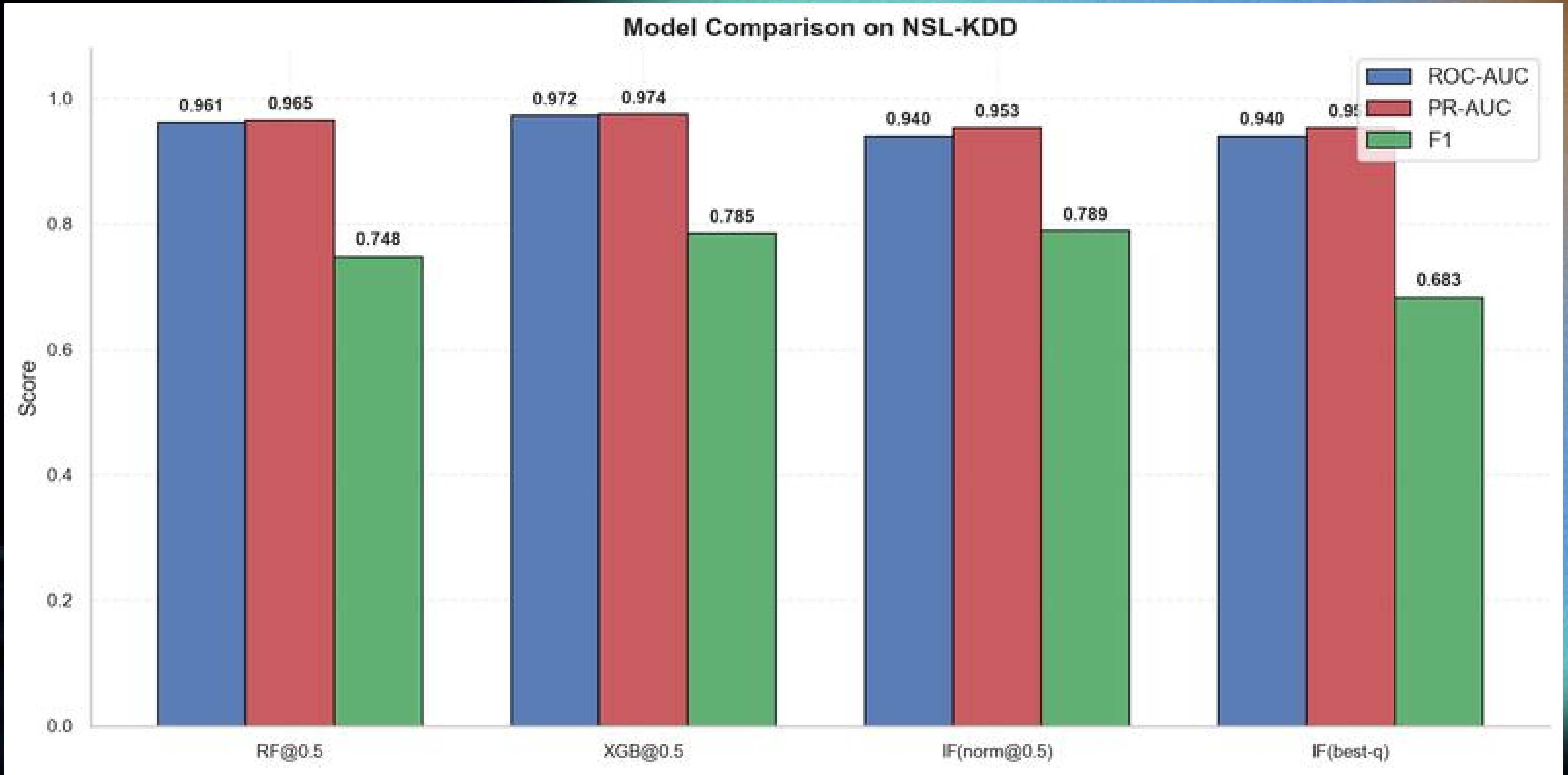
- Normal salary values are predicted as inliers (label = 1)
- Extremely large values such as 33B and 44B are predicted as outliers (label = -1)
- Main point:
  - IF detects clear outliers very well
  - this is an “easy case” for IF

NSL_KDD DATASET OVERVIEW



NSL-KDD is relatively more balanced and cleaner than the main dataset, so strong PR-AUC and F1 are easier to achieve here.

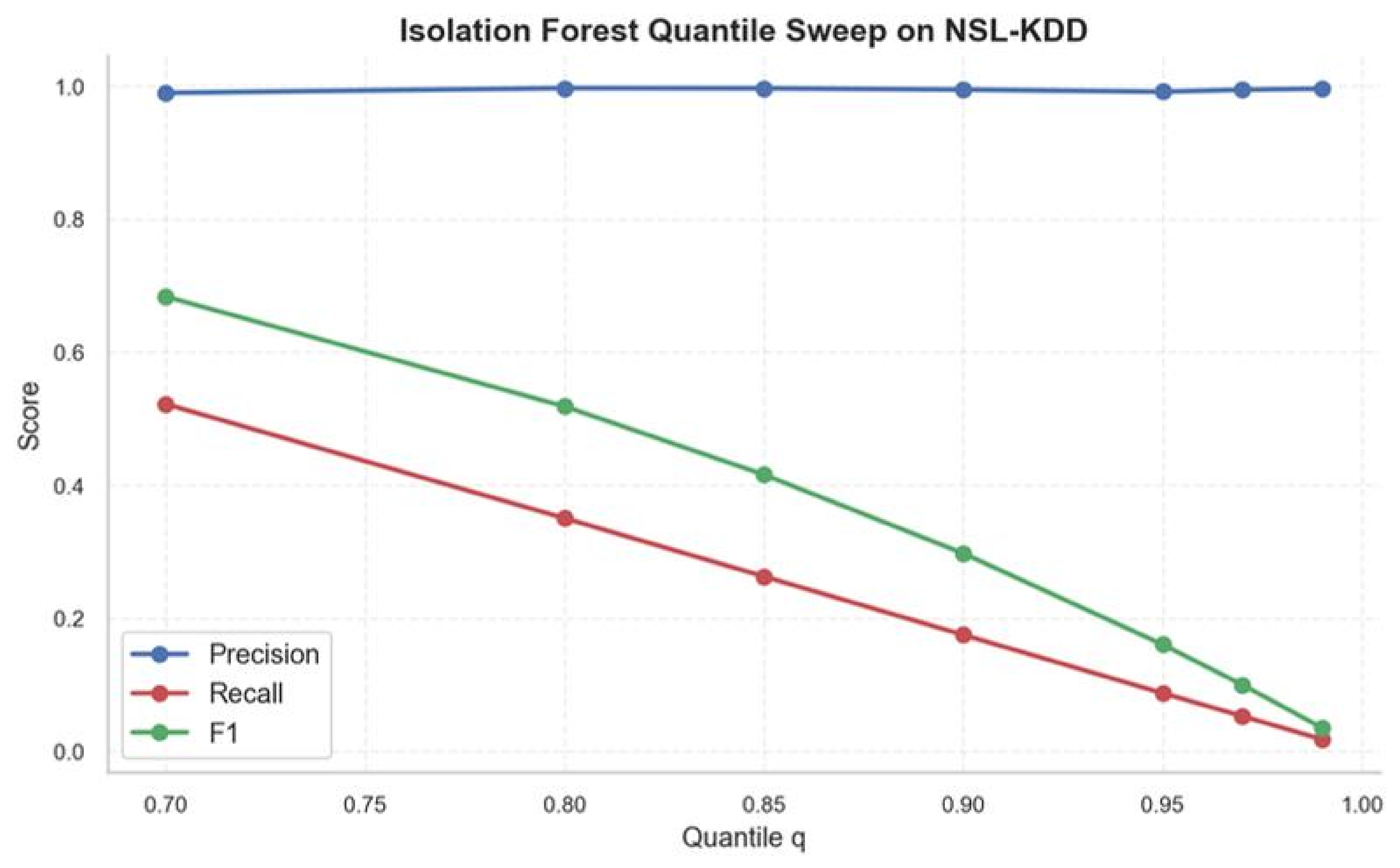
MODEL COMPARISON ON NSL-KDD



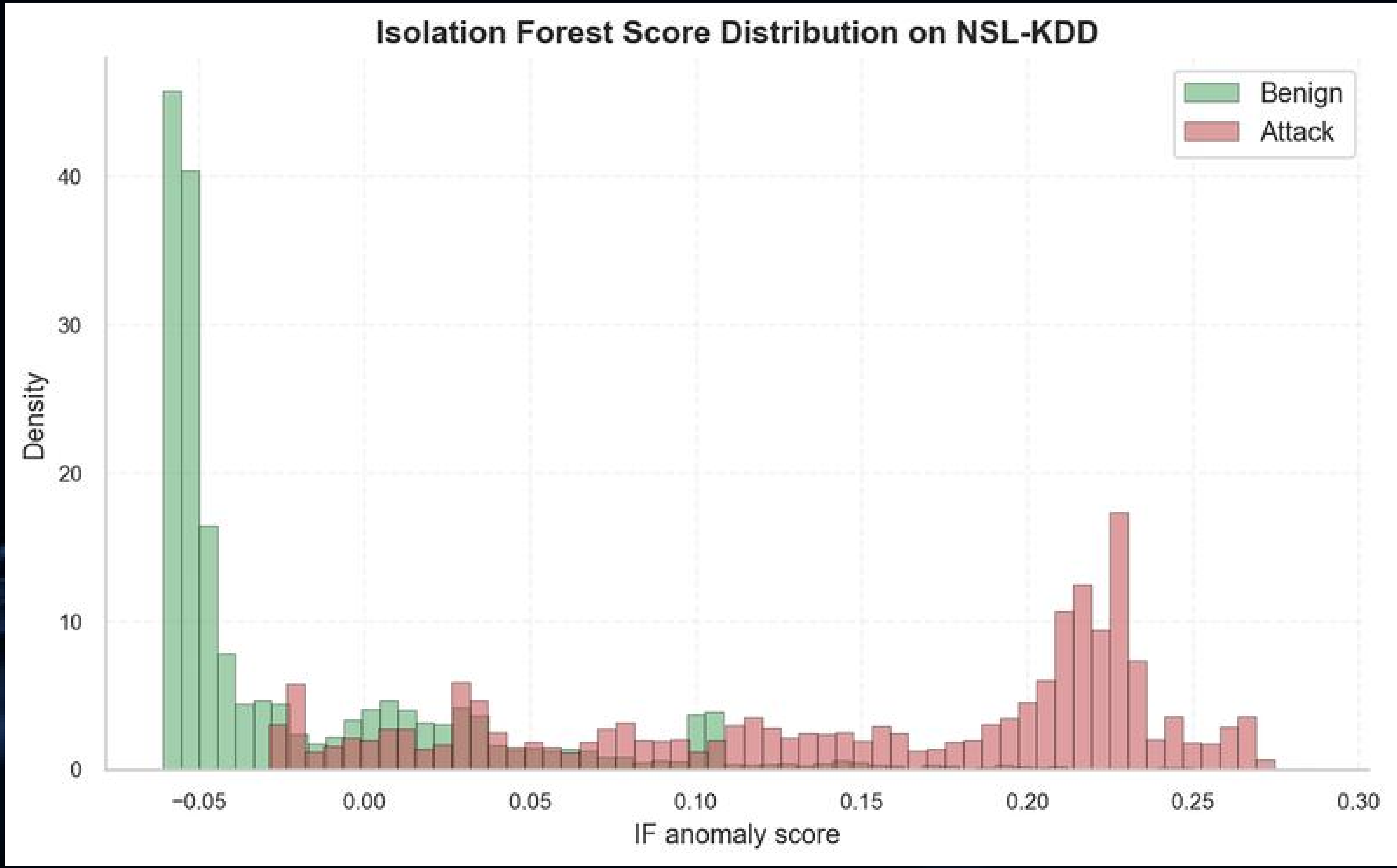
- XGB is the best supervised model overall
- RF is slightly behind
- IF is not weak on NSL-KDD
- Under the normalized threshold setting, IF reaches F1 = 0.789, which is close to XGB (0.785)

IF THRESHOLD SENSITIVITY

- Precision remains extremely high across quantiles
- Recall drops sharply as the threshold becomes stricter
- IF is highly sensitive to threshold selection
- At q = 0.70, IF achieves the best F1 (0.683)
- At q = 0.90, recall drops to about 0.175



WHY IF WORKS BETTER ON NSL-KDD



- Benign scores are concentrated around -0.06 to 0
- Attack scores are concentrated more around 0.10 to 0.25
- There is overlap around 0.00 to 0.10
- So IF can separate the two classes reasonably well, but not perfectly

This explains why IF performs better on NSL-KDD than on NF-UNSW-NB15.

WEAKNESSES OF IF ON REAL-WORLD NETFLOW

- IF works well only when attacks are far away from normal traffic in feature space.
- In real network traffic, many attack flows overlap with benign traffic in feature space.
- Therefore, IF may miss many malicious flows

RF WEAKNESSES

dst_port	protocol	avg_pkt_size	label
445	6	510	Attack
445	6	530	Attack
520	17	480	Attack
80	6	500	Benign
53	17	470	Benign
445	6	520	**Benign (new environment)**

Random Forest: Prone to learning "Shortcuts" (e.g., identifying labels based solely on Port/Protocol) instead of actual behavior.

XGB WEAKNESSES

max_ttl	iat_stddev	throughput	label
255	2	91	Attack
252	3	89	Attack
32	40	15	Benign
31	37	18	Benign
250	35	22	Benign (new domain)
40	5	88	Attack (new domain)

XGBoost: Strong at learning sharp decision boundaries, but more sensitive to feature shift and domain shift across datasets.

FEATURE ABLATION

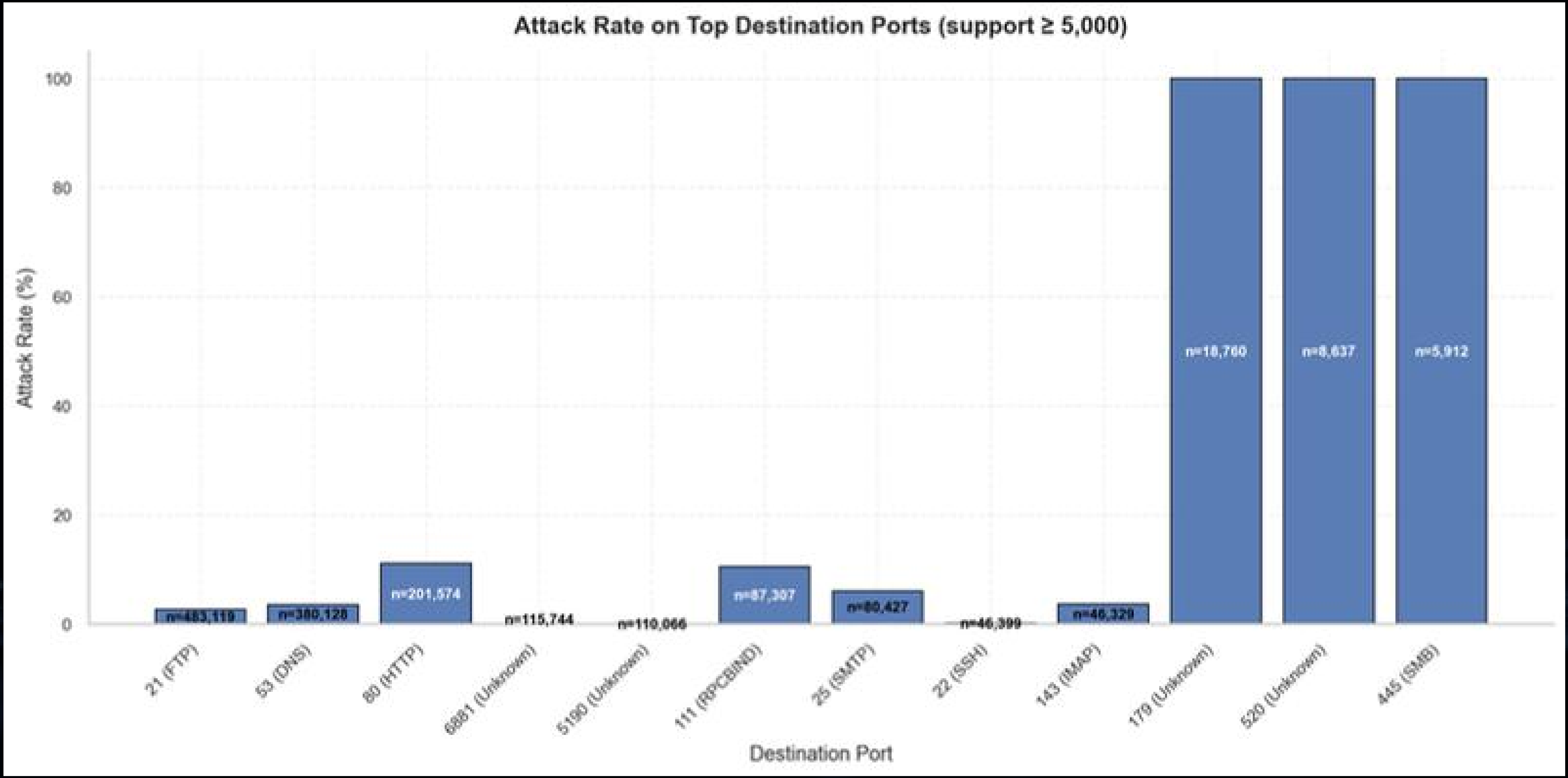
PURPOSE OF FEATURE ABLATION

- Check whether the models learn **real traffic behavior or just shortcuts**
- Reduce dependence on **overly strong features**
- Identify a final feature setting that balances:
    - high performance
    - better credibility
    - stronger generalization

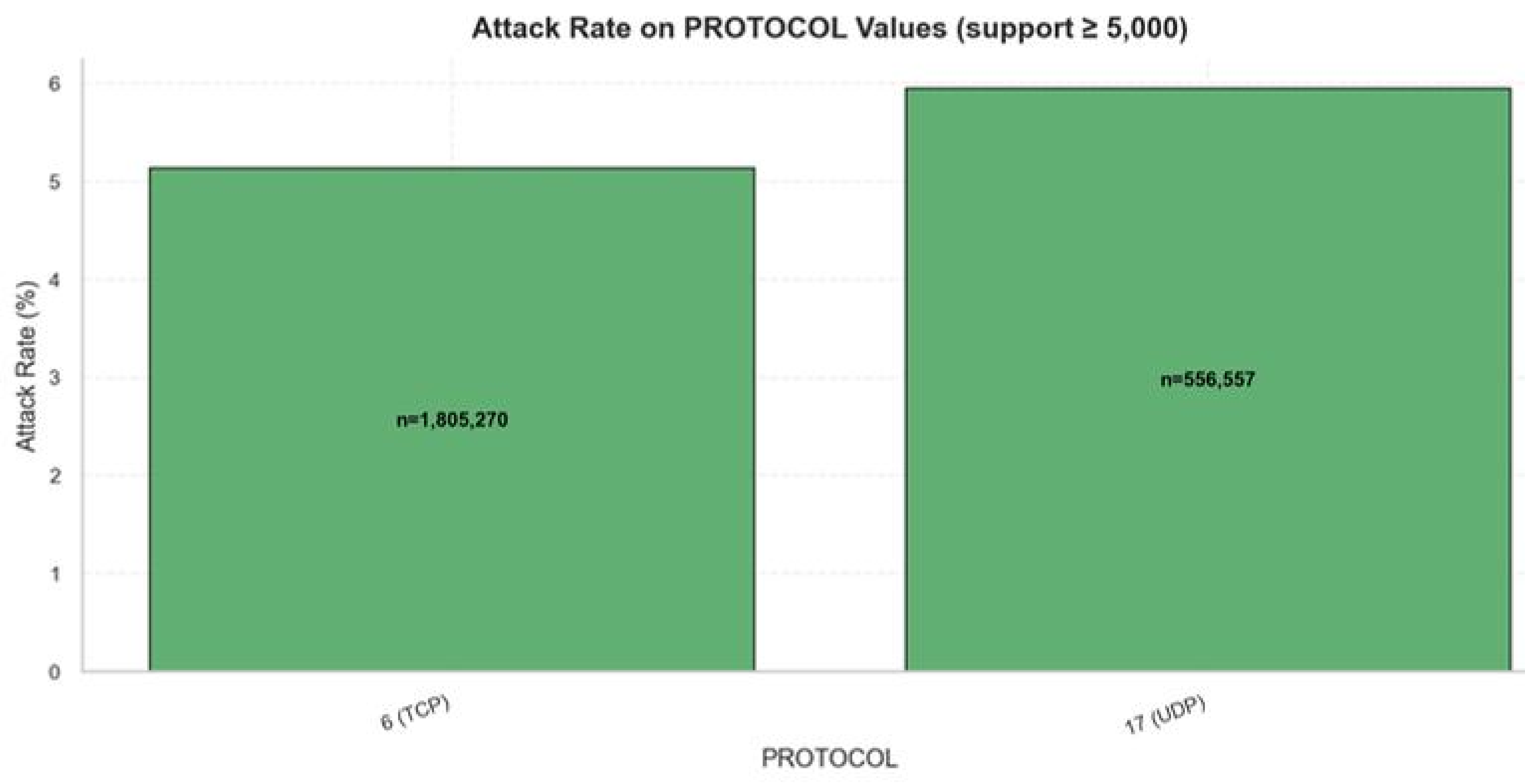
PROTOCOL/PORT BIAS

Main point:

- protocol and destination port can easily become shortcut features
- PROTOCOL explains traffic composition and mild label skew
- Destination ports provide much stronger shortcut signals
- Some ports such as 179, 520, 445 have attack rates near 100%



Attack Rate on Top Destination Ports (support ≥ 5,000)
Attack Rate (%)
100
80
60
40
20
0
n=483,119
n=380,128
n=201,574
n=115,744
n=110,066
n=87,307
n=80,427
n=46,399
n=46,329
n=18,760
n=8,637
n=5,912
21 (FTP)
53 (DNS)
80 (HTTP)
6881 (Unknown)
5190 (Unknown)
111 (RPCBIND)
25 (SMTP)
22 (SSH)
143 (IMAP)
179 (Unknown)
520 (Unknown)
445 (SMB)
Destination Port



Attack Rate on PROTOCOL Values (support ≥ 5,000)
Attack Rate (%)
6
5
4
3
2
1
0
n=1,805,270
n=556,557
6 (TCP)
17 (UDP)
PROTOCOL

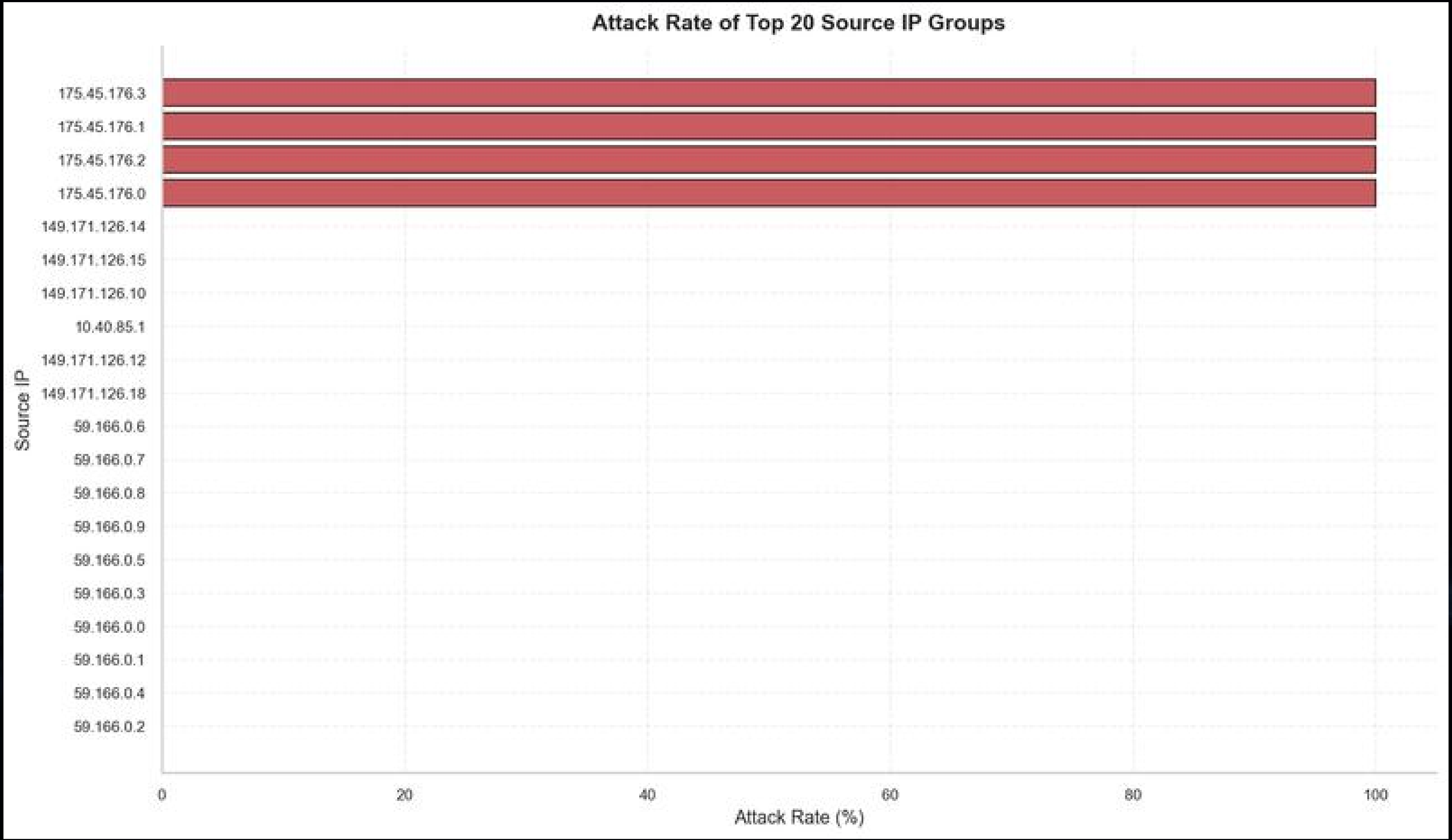
SOURCE-GROUP DEPENDENCE

Main point:

- Some source IP groups have extremely concentrated attack behavior, reaching nearly 100% attack rate.

This explains:

- why we split by IPV4_SRC_ADDR
- why behavior-related features must be treated carefully


Attack Rate of Top 20 Source IP Groups
175.45.176.3
175.45.176.1
175.45.176.2
175.45.176.0
149.171.126.14
149.171.126.15
149.171.126.10
10.40.85.1
149.171.126.12
149.171.126.18
59.166.0.6
59.166.0.7
59.166.0.8
59.166.0.9
59.166.0.5
59.166.0.3
59.166.0.0
59.166.0.1
59.166.0.4
59.166.0.2
Source IP
0
20
40
60
80
100
Attack Rate (%)

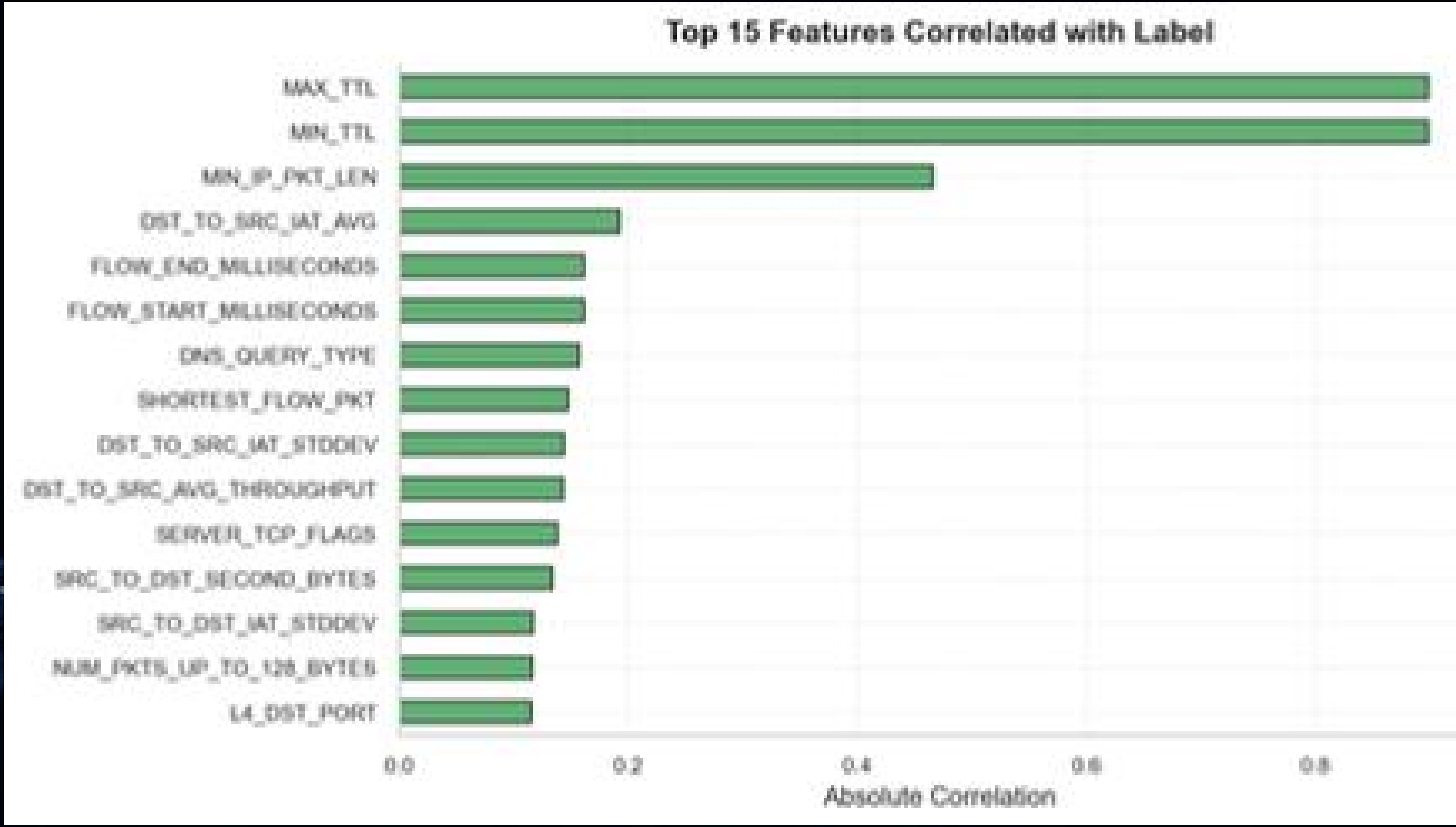
FEATURE REDUNDANCY

Important numbers:

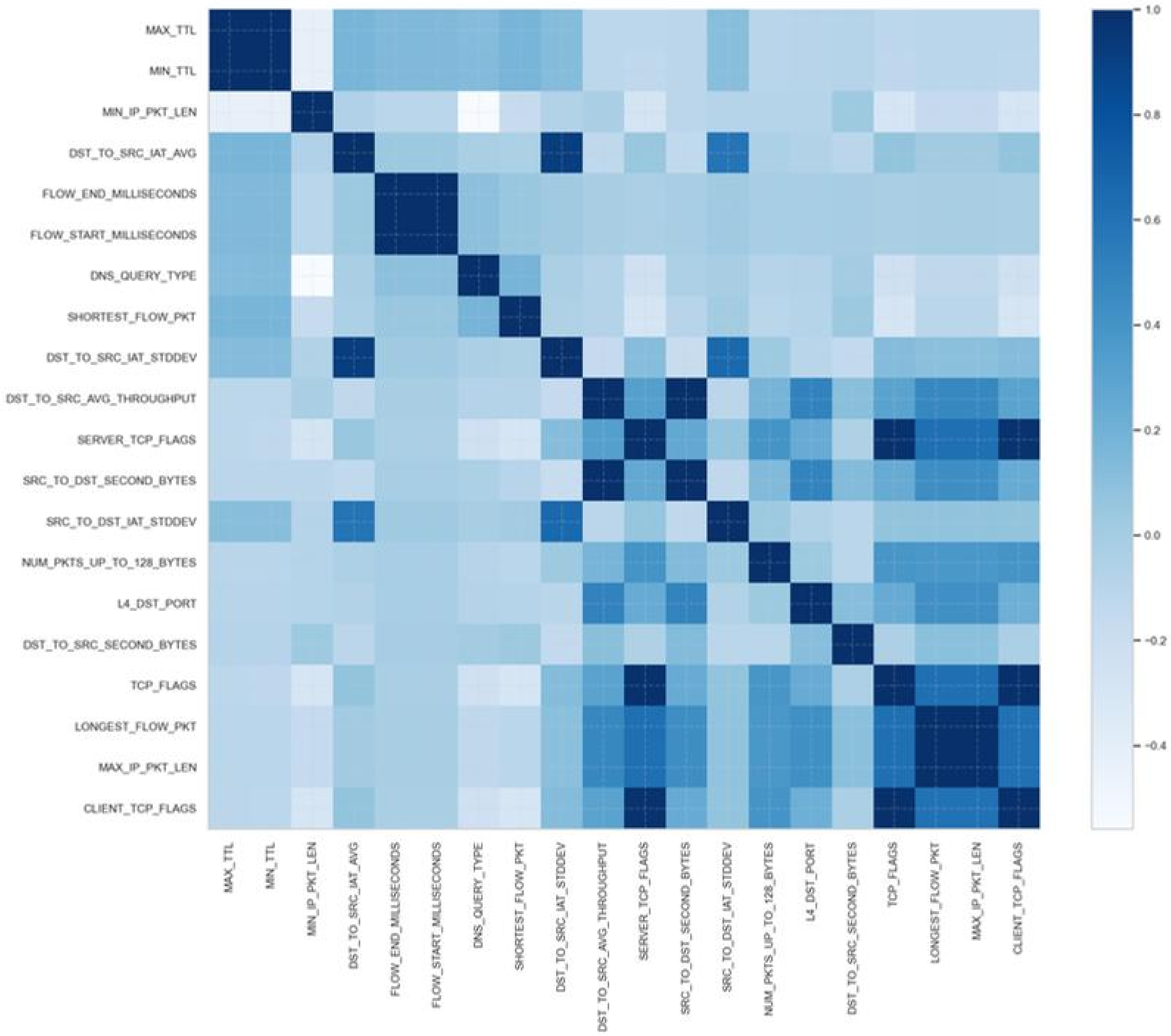
- MAX_TTL and MIN_TTL correlation with label: about 0.898
- MIN_IP_PKT_LEN correlation with label: about 0.466

Main point:

- many features are correlated with both the label and each other
- therefore, removing a single feature group does not immediately collapse model performance


Top 15 Features Correlated with Label
MAX_TTL
MIN_TTL
MIN_IP_PKT_LEN
DST_TO_SRC_IAT_AVG
FLOW_END_MILLISECONDS
FLOW_START_MILLISECONDS
DNS_QUERY_TYPE
SHORTEST_FLOW_PKT
DST_TO_SRC_IAT_STDDEV
DST_TO_SRC_AVG_THROUGHPUT
SERVER_TCP_FLAGS
SRC_TO_DST_SECOND_BYTES
SRC_TO_DST_IAT_STDDEV
NUM_PKTS_UP_TO_128_BYTES
L4_DST_PORT
0.0
0.2
0.4
0.6
0.8
Absolute Correlation

Correlation Heatmap for Top 20 Label-Related Features



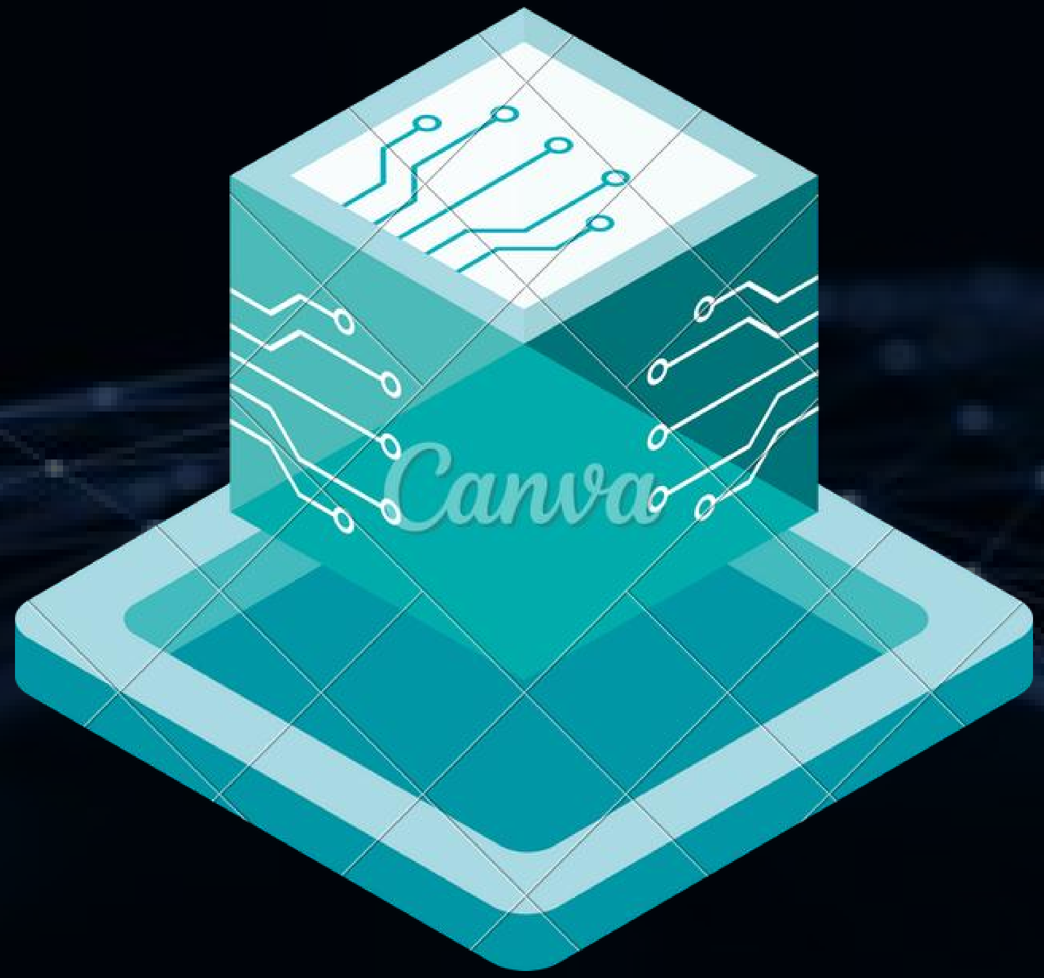
MODEL LIMITATION SUMMARY

Why No Model Is Enough Alone

- IF → fails with overlapping distributions
- RF → learns shortcuts
- XGB → sensitive to feature/domain shift

“We need better feature design, not just better models”

ABLATION DESIGN & RESULTS



Experimental Configurations:

- Raw Feature Baseline.
- Full Enriched Feature Set.
- Without Protocol/Port and Behavioral Semantics.
- Selective Feature Reduction (Final Setting).

Key Finding:

- Removing individual feature groups does not collapse performance; the final setting provides a more robust and defensible configuration.

FINAL FEATURE CONFIGURATION

Main findings:

- dropping a single feature group does not immediately destroy performance
- this confirms strong feature redundancy
- the final setting:
  - reduces feature count from 68 to 27
  - still keeps strong performance

Key message:

- the final setting was chosen not only for strong scores, but also because it is more defensible and less shortcut-driven.

FINAL FEATURE ABLATION EXPERIMENTAL RESULTS

FINAL FEATURE ABLATION EXPERIMENTAL RESULTS

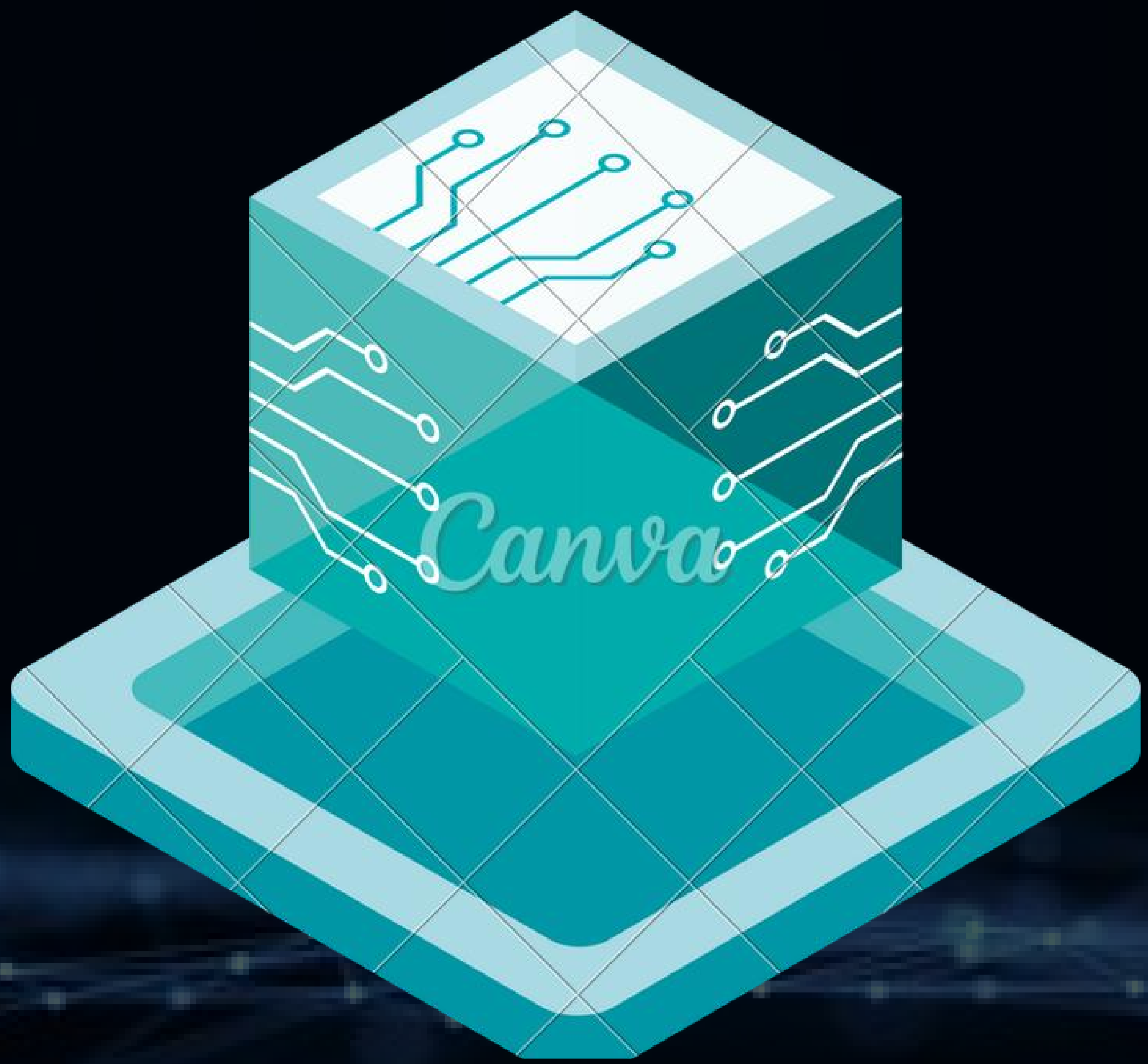
- Dataset: V3
- Pipeline: Group-split behavioral feature set
- Number of features: **27**
- Train size: **1,886,156**
- Test size: **413,167**
- Positive rate:
  - Train: **5.3250%**
  - Test: **6.4799%**

BEFORE VS AFTER ABLATION

FEATURE REDUCTION: 68 → 27

All models trained on final 27-feature configuration

- Accuracy / F1: Largely Maintained
- Model Behavior: Reduced Shortcut Sependency
- Generalization: Improved (Link Domain Shift)



FINAL RESULTS (AFTER FEATURE ABLATION)

All models trained on final 27-feature configuration

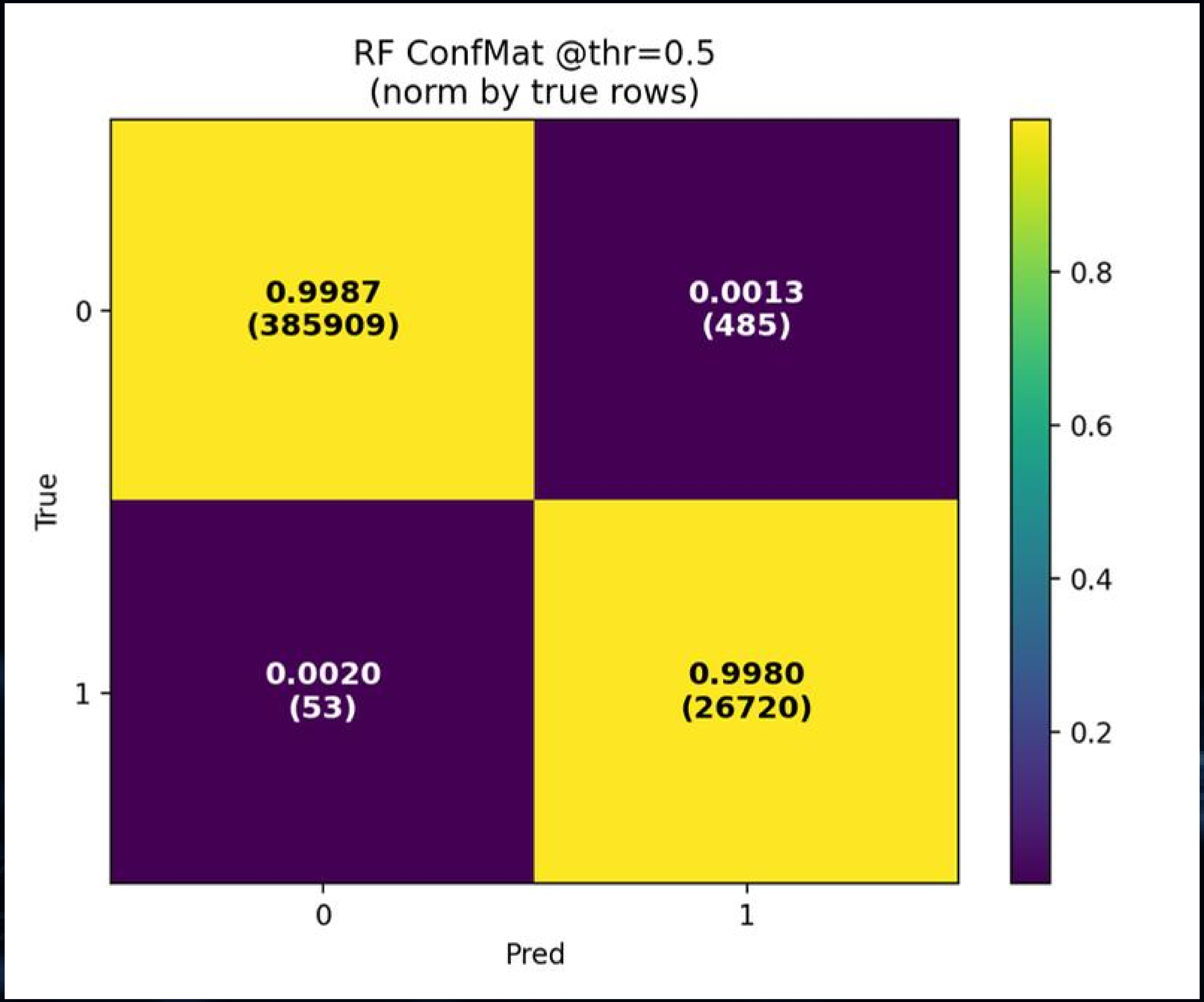
RF RESULTS @ THRESHOLD 0.5

Random Forest

- Precision: 0.9822
- Recall: 0.9980
- F1: 0.9900
- ROC-AUC: 0.999754
- PR-AUC: 0.995131

- TN: 385,909
- FP: 485
- FN: 53
- TP: 26,720

RF:
→ “High recall → catches most attacks”



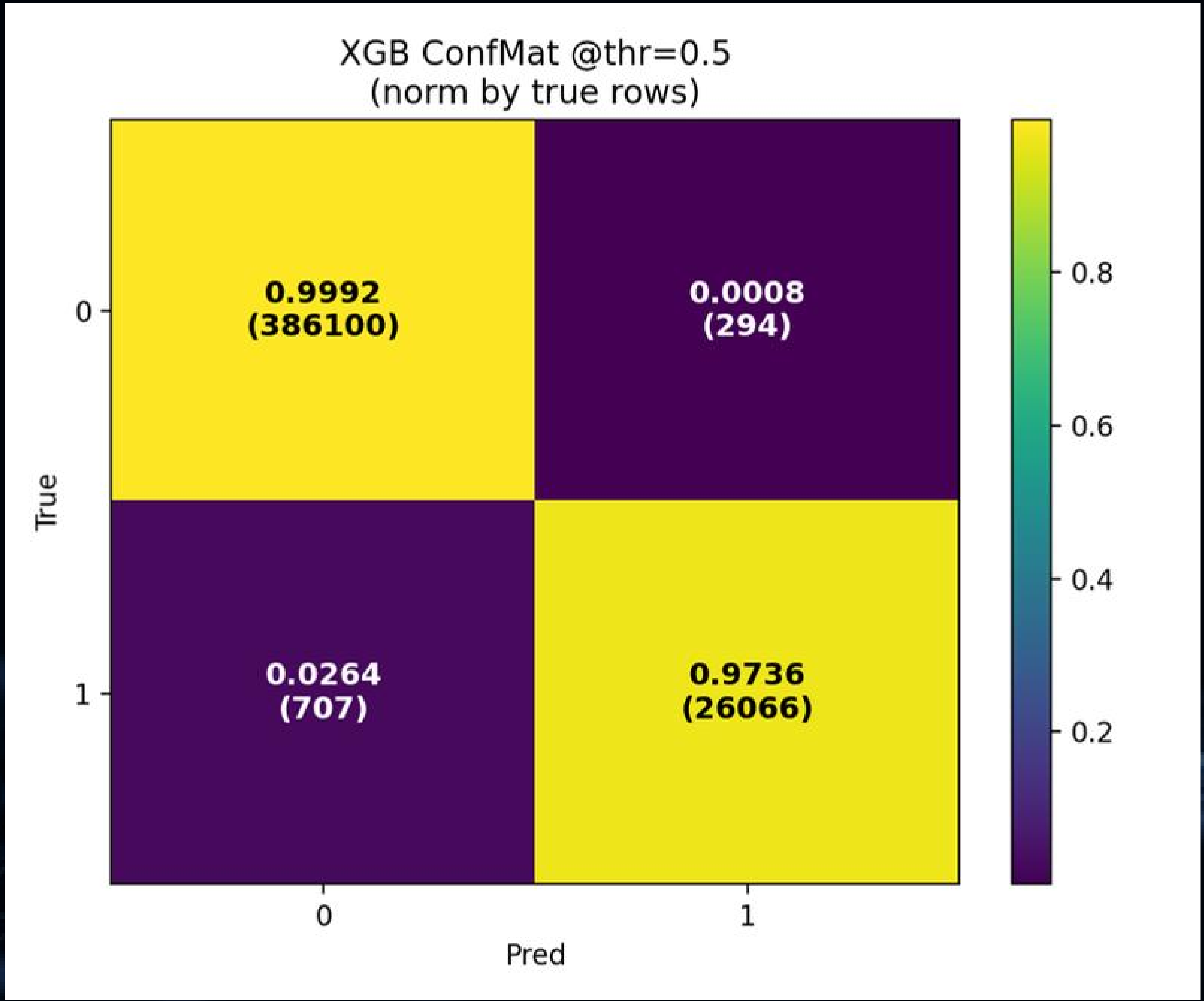
XGB RESULTS @ THRESHOLD 0.5

XGBoost

- Precision: 0.9888
- Recall: 0.9736
- F1: 0.9812
- ROC-AUC: 0.999778
- PR-AUC: 0.996806

- TN: 386,100
- FP: 294
- FN: 707
- TP: 26,066

XGB:
→ “Better ranking → fewer false positives”



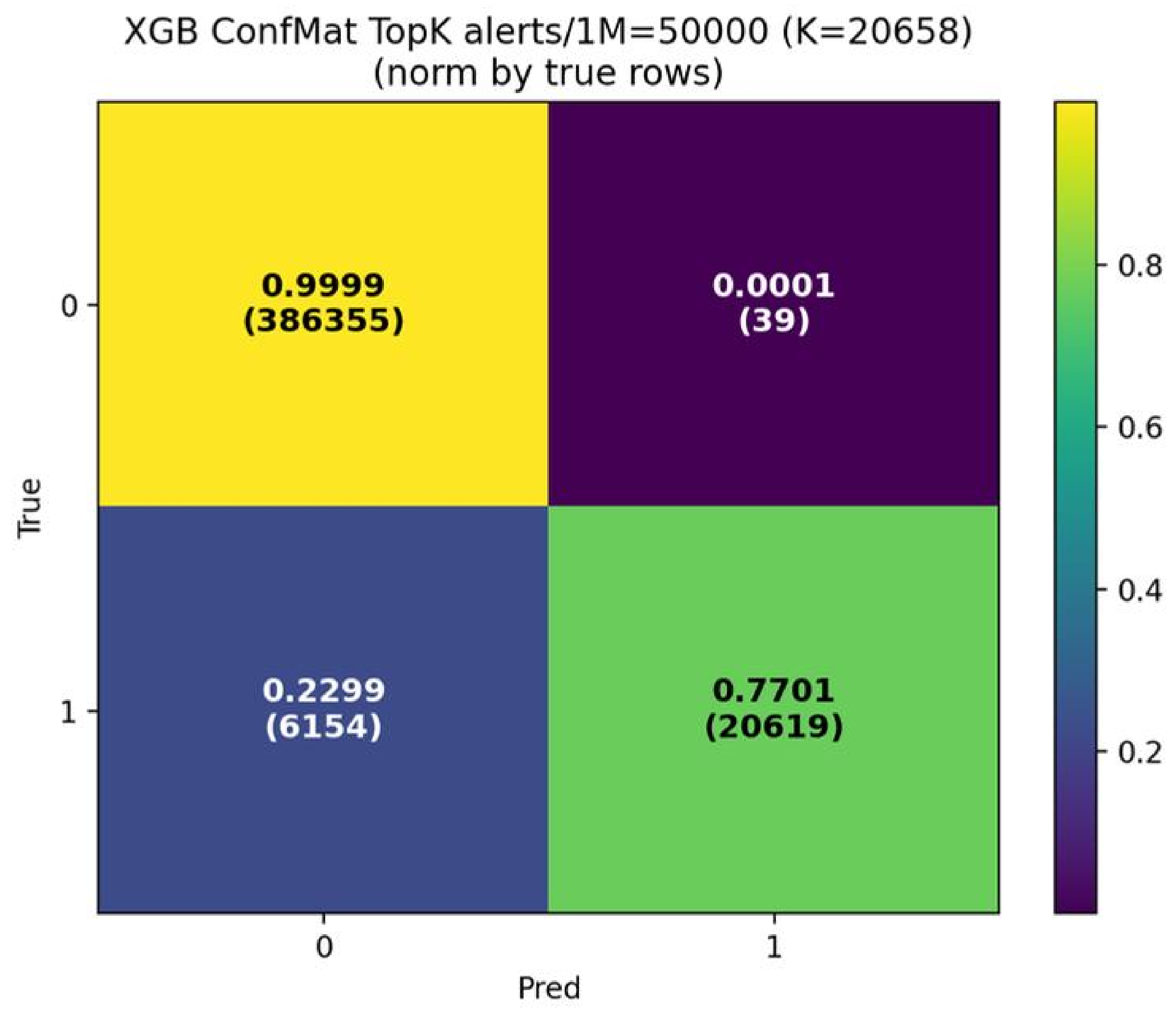
RESULTS UNDER ALERT BUDGET

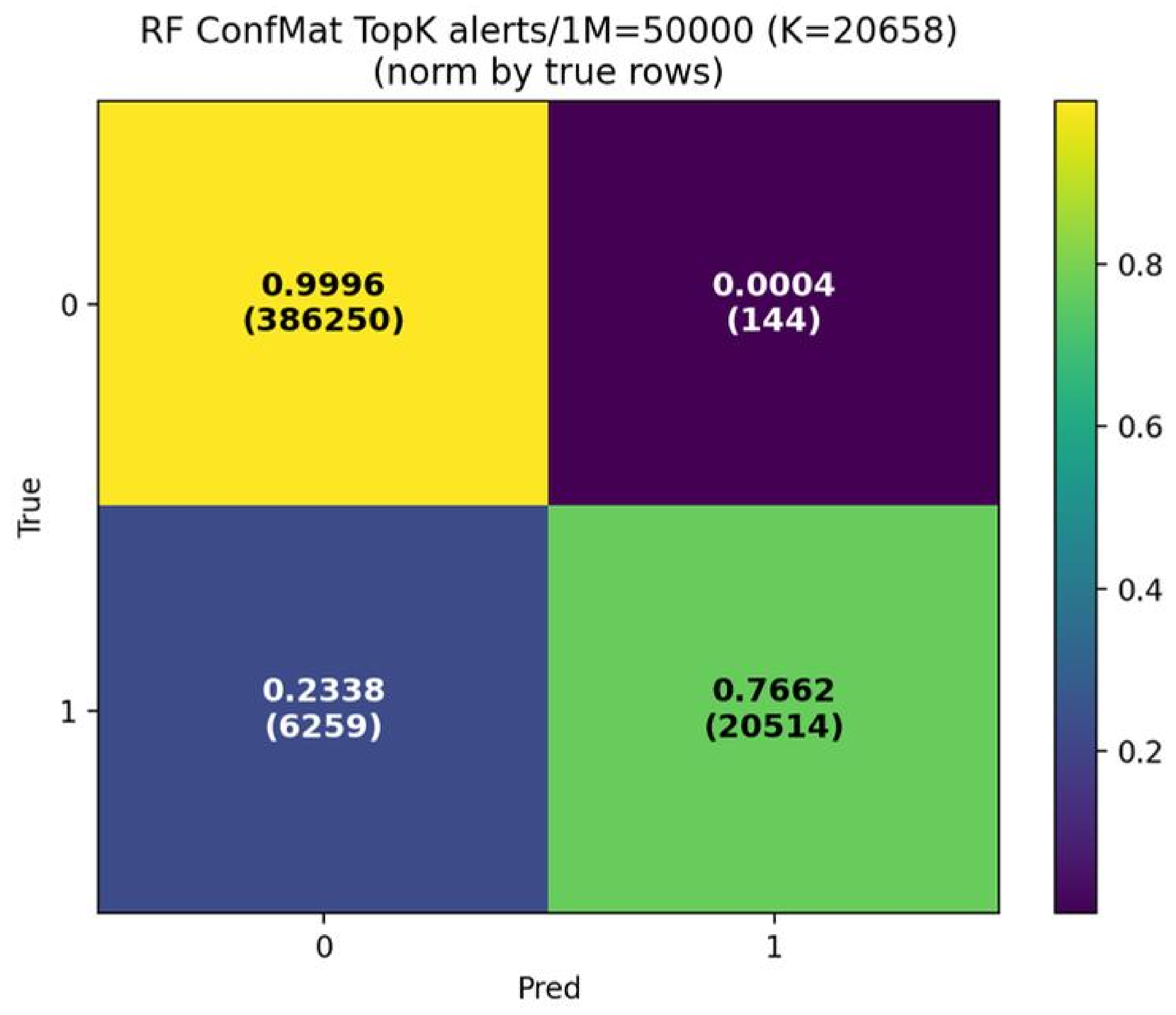
At 50,000 alerts / 1M flows(≈ 5% flows → simulates limited SOC analyst capacity)

- RF Precision@K: 0.9932
- RF Recall@K: 0.7662
- XGB Precision@K: 0.9988
- XGB Recall@K: 0.7701

Key message:

- both supervised models remain strong
- XGB is slightly better under a fixed alert budget





IF UNDER ALERT BUDGET

Isolation Forest

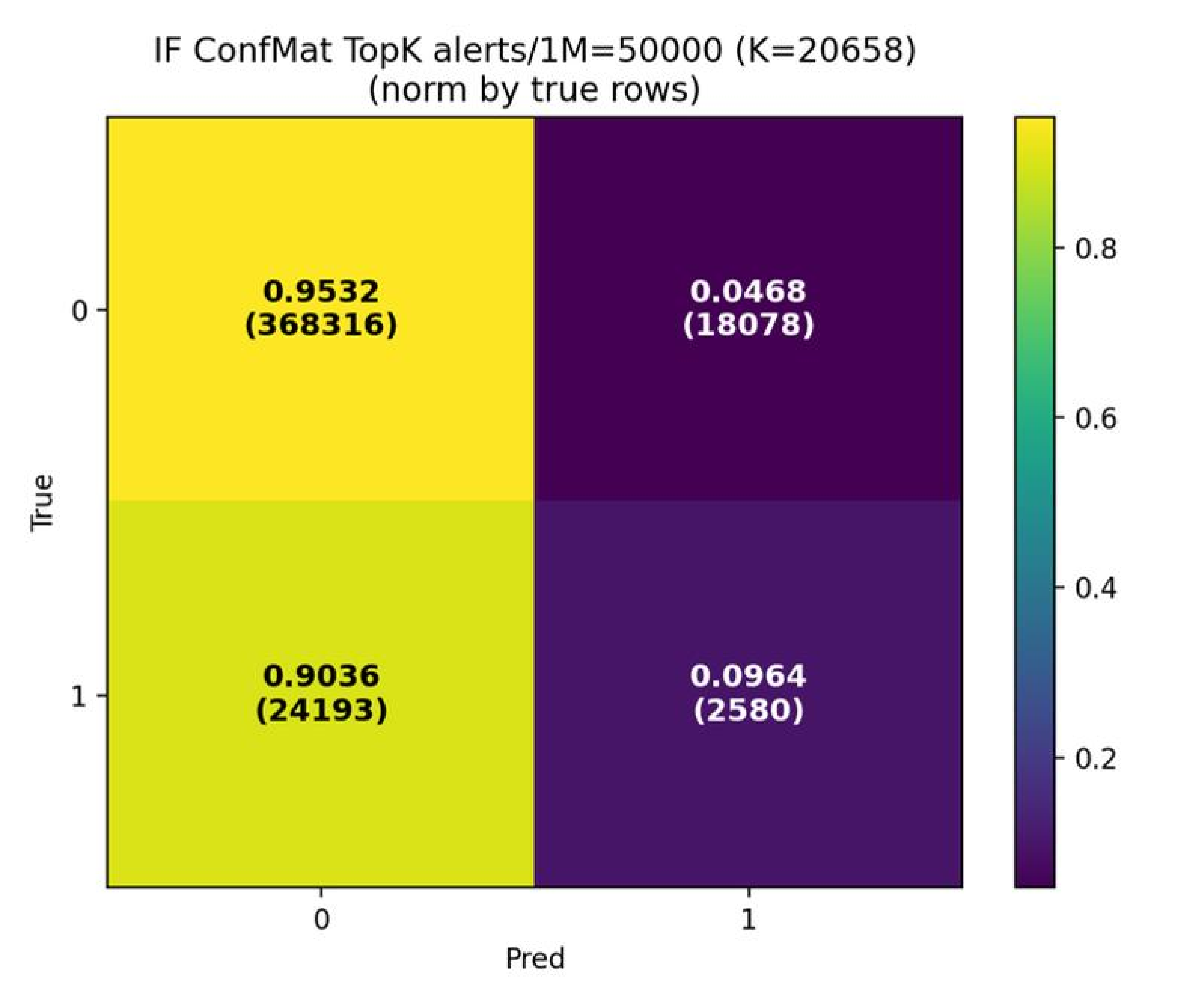
- ROC-AUC: about 0.6854
- PR-AUC: about 0.1160
- Best F1: about 0.1837

At 50,000 alerts / 1M flows(≈ 5% flows):

- Precision@K: 0.1249
- Recall@K: 0.0964

Key message:

- IF performs much worse than RF and XGB
- IF is better treated as an unsupervised baseline



FINAL RESULTS SUMMARY

- RF is best by F1 @ 0.5
- XGB is very strong in ranking / Top-K alerting
- IF serves as the unsupervised baseline
- The final feature set remains very strong even after reducing from 68 to 27 features

DOMAIN SHIFT & CROSS-DATASET EVALUATION

DOMAIN SHIFT ANALYSIS

Why?

- Strong in-domain performance is not enough
- We must check whether the models generalize between V2 and V3

Result

- **After sampling:**
- V2: 1,000,000 samples
- V3: 1,000,000 samples
- Common numeric features kept: 33
- Domain-classifier ROC-AUC = 1.0000

Finding

- There is strong domain shift between V2 and V3

MEANING OF THE DOMAIN SHIFT

```
=== INTERPRETATION ===
Domain-classifier ROC-AUC = 1.0000
=> Strong domain shift.
Features with KS p < 1e-6 : 33
Features with Wasserstein_z > 0.5 : 2
Features with PSI > 0.2 : 8
```

- V2 and V3 differ clearly in feature distribution
- Therefore:
  - high in-domain performance does not guarantee strong cross-dataset performance
  - train-on-one, test-on-another results may drop significantly
- “Model trained on V3 behaves like it sees a completely different world when tested on V2”

TRAIN V2 → TEST V2

Scenario	Model	K	TP	FP	TN	FN	Recall@K	Precision@K	F1_Score	Max_Possible_Recall	Efficiency (%)	Alerts/1M_flows
Internal_TrainV2_TestV2	Random Forest	50	50	0	459044	18961	0.0026	1	0.0052	0.0026	100	104.59
Internal_TrainV2_TestV2	Random Forest	100	100	0	459044	18911	0.0053	1	0.0105	0.0053	100	209.18
Internal_TrainV2_TestV2	Random Forest	500	500	0	459044	18511	0.0263	1	0.0513	0.0263	100	1045.9
Internal_TrainV2_TestV2	Random Forest	1000	1000	0	459044	18011	0.0526	1	0.0999	0.0526	100	2091.81
Internal_TrainV2_TestV2	Random Forest	5000	4996	4	459040	14015	0.2628	0.9992	0.4161	0.263	99.92	10459.05
Internal_TrainV2_TestV2	Random Forest	10000	9881	119	458925	9130	0.5198	0.9881	0.6812	0.526	98.81	20918.1
Internal_TrainV2_TestV2	Random Forest	20000	18570	1430	457614	441	0.9768	0.9285	0.952	1	97.68	41836.19
Internal_TrainV2_TestV2	Random Forest	30000	19001	10999	448045	10	0.9995	0.6334	0.7754	1	99.95	62754.29
Internal_TrainV2_TestV2	XGBoost	50	49	1	459043	18962	0.0026	0.98	0.0051	0.0026	98	104.59
Internal_TrainV2_TestV2	XGBoost	100	99	1	459043	18912	0.0052	0.99	0.0104	0.0053	99	209.18
Internal_TrainV2_TestV2	XGBoost	500	498	2	459042	18513	0.0262	0.996	0.051	0.0263	99.6	1045.9
Internal_TrainV2_TestV2	XGBoost	1000	998	2	459042	18013	0.0525	0.998	0.0997	0.0526	99.8	2091.81
Internal_TrainV2_TestV2	XGBoost	5000	4982	18	459026	14029	0.2621	0.9964	0.415	0.263	99.64	10459.05
Internal_TrainV2_TestV2	XGBoost	10000	9884	116	458928	9127	0.5199	0.9884	0.6814	0.526	98.84	20918.1
Internal_TrainV2_TestV2	XGBoost	20000	18476	1524	457520	535	0.9719	0.9238	0.9472	1	97.19	41836.19
Internal_TrainV2_TestV2	XGBoost	30000	19000	11000	448044	11	0.9994	0.6333	0.7753	1	99.94	62754.29
Internal_TrainV2_TestV2	Isolation Forest	50	37	13	459031	18974	0.0019	0.74	0.0039	0.0026	74	104.59
Internal_TrainV2_TestV2	Isolation Forest	100	77	23	459021	18934	0.0041	0.77	0.0081	0.0053	77	209.18
Internal_TrainV2_TestV2	Isolation Forest	500	390	110	458934	18621	0.0205	0.78	0.04	0.0263	78	1045.9
Internal_TrainV2_TestV2	Isolation Forest	1000	780	220	458824	18231	0.041	0.78	0.078	0.0526	78	2091.81
Internal_TrainV2_TestV2	Isolation Forest	5000	3312	1688	457356	15699	0.1742	0.6624	0.2759	0.263	66.24	10459.05
Internal_TrainV2_TestV2	Isolation Forest	10000	6723	3277	455767	12288	0.3536	0.6723	0.4635	0.526	67.23	20918.1
Internal_TrainV2_TestV2	Isolation Forest	20000	11943	8057	450987	7068	0.6282	0.5972	0.6123	1	62.82	41836.19
Internal_TrainV2_TestV2	Isolation Forest	30000	16095	13905	445139	2916	0.8466	0.5365	0.6568	1	84.66	62754.29

['OUT_BYTES', 'DST_TO_SRC_SECOND_BYTES', 'SERVER_TCP_FLAGS', 'OUT_PKTS', 'DST_TO_SRC_AVG_THROUGHPUT', 'NUM_PKTS_UP_TO_128_BYTES', 'TCP_FLAGS', 'SRC_TO_DST_SECOND_BYTES', 'IN_PKTS', 'CLIENT_TCP_FLAGS', 'RETRANSMITTED_IN_BYTES', 'ICMP_IPV4_TYPE', 'IN_BYTES', 'MAX_IP_PKT_LEN', 'LONGEST_FLOW_PKT']

TRAIN V3 → TEST V3

Scenario	Model	K	TP	FP	TN	FN	Recall@K	Precision@K	F1_Score	Max_Possible_Recall	Efficiency (%)	Alerts/1M_flows
Internal_TrainV3_TestV3	Random Forest	50	50	0	447546	25489	0.002	1	0.0039	0.002	100	105.69
Internal_TrainV3_TestV3	Random Forest	100	100	0	447546	25439	0.0039	1	0.0078	0.0039	100	211.38
Internal_TrainV3_TestV3	Random Forest	500	500	0	447546	25039	0.0196	1	0.0384	0.0196	100	1056.89
Internal_TrainV3_TestV3	Random Forest	1000	1000	0	447546	24539	0.0392	1	0.0754	0.0392	100	2113.79
Internal_TrainV3_TestV3	Random Forest	5000	5000	0	447546	20539	0.1958	1	0.3275	0.1958	100	10568.93
Internal_TrainV3_TestV3	Random Forest	10000	9998	2	447544	15541	0.3915	0.9998	0.5626	0.3916	99.98	21137.85
Internal_TrainV3_TestV3	Random Forest	20000	19932	68	447478	5607	0.7805	0.9966	0.8754	0.7831	99.66	42275.7
Internal_TrainV3_TestV3	Random Forest	30000	25539	4461	443085	0	1	0.8513	0.9197	1	100	63413.55
Internal_TrainV3_TestV3	XGBoost	50	50	0	447546	25489	0.002	1	0.0039	0.002	100	105.69
Internal_TrainV3_TestV3	XGBoost	100	100	0	447546	25439	0.0039	1	0.0078	0.0039	100	211.38
Internal_TrainV3_TestV3	XGBoost	500	499	1	447545	25040	0.0195	0.998	0.0383	0.0196	99.8	1056.89
Internal_TrainV3_TestV3	XGBoost	1000	999	1	447545	24540	0.0391	0.999	0.0753	0.0392	99.9	2113.79
Internal_TrainV3_TestV3	XGBoost	5000	4999	1	447545	20540	0.1957	0.9998	0.3274	0.1958	99.98	10568.93
Internal_TrainV3_TestV3	XGBoost	10000	9995	5	447541	15544	0.3914	0.9995	0.5625	0.3916	99.95	21137.85
Internal_TrainV3_TestV3	XGBoost	20000	19983	17	447529	5556	0.7825	0.9992	0.8776	0.7831	99.92	42275.7
Internal_TrainV3_TestV3	XGBoost	30000	25537	4463	443083	2	0.9999	0.8512	0.9196	1	99.99	63413.55
Internal_TrainV3_TestV3	Isolation Forest	50	46	4	447542	25493	0.0018	0.92	0.0036	0.002	92	105.69
Internal_TrainV3_TestV3	Isolation Forest	100	91	9	447537	25448	0.0036	0.91	0.0071	0.0039	91	211.38
Internal_TrainV3_TestV3	Isolation Forest	500	461	39	447507	25078	0.0181	0.922	0.0354	0.0196	92.2	1056.89
Internal_TrainV3_TestV3	Isolation Forest	1000	947	53	447493	24592	0.0371	0.947	0.0714	0.0392	94.7	2113.79
Internal_TrainV3_TestV3	Isolation Forest	5000	4482	518	447028	21057	0.1755	0.8964	0.2935	0.1958	89.64	10568.93
Internal_TrainV3_TestV3	Isolation Forest	10000	8897	1103	446443	16642	0.3484	0.8897	0.5007	0.3916	88.97	21137.85
Internal_TrainV3_TestV3	Isolation Forest	20000	16332	3668	443878	9207	0.6395	0.8166	0.7173	0.7831	81.66	42275.7
Internal_TrainV3_TestV3	Isolation Forest	30000	22551	7449	440097	2988	0.883	0.7517	0.8121	1	88.3	63413.55

['OUT_BYTES', 'DST_TO_SRC_SECOND_BYTES', 'SERVER_TCP_FLAGS', 'OUT_PKTS', 'DST_TO_SRC_AVG_THROUGHPUT', 'NUM_PKTS_UP_TO_128_BYTES', 'TCP_FLAGS', 'SRC_TO_DST_SECOND_BYTES', 'IN_PKTS', 'CLIENT_TCP_FLAGS', 'RETRANSMITTED_IN_BYTES', 'ICMP_IPV4_TYPE', 'IN_BYTES', 'MAX_IP_PKT_LEN', 'LONGEST_FLOW_PKT']

CROSS-DATASET EVALUATION SETUP

Scenarios:

Train on V2 --> Test on V3.
Train on V3 --> Test on V2.

Approach:

Utilizing the refined 15-feature set.

TRAIN V2 → TEST V3

Scenario	Model	K	TP	FP	TN	FN	Recall@K	Precision@K	F1_Score	Max_Possible_Recall	Efficiency (%)	Alerts/1M_flows
CrossEval_TrainV2_TestV3_Full	Random Forest	50	50	0	2237731	127643	0.0004	1	0.0008	0.0004	100	21.14
CrossEval_TrainV2_TestV3_Full	Random Forest	100	100	0	2237731	127593	0.0008	1	0.0016	0.0008	100	42.28
CrossEval_TrainV2_TestV3_Full	Random Forest	500	500	0	2237731	127193	0.0039	1	0.0078	0.0039	100	211.38
CrossEval_TrainV2_TestV3_Full	Random Forest	1000	1000	0	2237731	126693	0.0078	1	0.0155	0.0078	100	422.76
CrossEval_TrainV2_TestV3_Full	Random Forest	5000	4993	7	2237724	122700	0.0391	0.9986	0.0753	0.0392	99.86	2113.79
CrossEval_TrainV2_TestV3_Full	Random Forest	10000	9993	7	2237724	117700	0.0783	0.9993	0.1451	0.0783	99.93	4227.57
CrossEval_TrainV2_TestV3_Full	Random Forest	20000	19969	31	2237700	107724	0.1564	0.9984	0.2704	0.1566	99.84	8455.14
CrossEval_TrainV2_TestV3_Full	Random Forest	30000	29633	367	2237364	98060	0.2321	0.9878	0.3758	0.2349	98.78	12682.72
CrossEval_TrainV2_TestV3_Full	XGBoost	50	50	0	2237731	127643	0.0004	1	0.0008	0.0004	100	21.14
CrossEval_TrainV2_TestV3_Full	XGBoost	100	100	0	2237731	127593	0.0008	1	0.0016	0.0008	100	42.28
CrossEval_TrainV2_TestV3_Full	XGBoost	500	500	0	2237731	127193	0.0039	1	0.0078	0.0039	100	211.38
CrossEval_TrainV2_TestV3_Full	XGBoost	1000	1000	0	2237731	126693	0.0078	1	0.0155	0.0078	100	422.76
CrossEval_TrainV2_TestV3_Full	XGBoost	5000	5000	0	2237731	122693	0.0392	1	0.0754	0.0392	100	2113.79
CrossEval_TrainV2_TestV3_Full	XGBoost	10000	10000	0	2237731	117693	0.0783	1	0.1453	0.0783	100	4227.57
CrossEval_TrainV2_TestV3_Full	XGBoost	20000	19949	51	2237680	107744	0.1562	0.9974	0.2701	0.1566	99.74	8455.14
CrossEval_TrainV2_TestV3_Full	XGBoost	30000	29401	599	2237132	98292	0.2302	0.98	0.3729	0.2349	98	12682.72
CrossEval_TrainV2_TestV3_Full	Isolation Forest	50	49	1	2237730	127644	0.0004	0.98	0.0008	0.0004	98	21.14
CrossEval_TrainV2_TestV3_Full	Isolation Forest	100	98	2	2237729	127595	0.0008	0.98	0.0015	0.0008	98	42.28
CrossEval_TrainV2_TestV3_Full	Isolation Forest	500	450	50	2237681	127243	0.0035	0.9	0.007	0.0039	90	211.38
CrossEval_TrainV2_TestV3_Full	Isolation Forest	1000	903	97	2237634	126790	0.0071	0.903	0.014	0.0078	90.3	422.76
CrossEval_TrainV2_TestV3_Full	Isolation Forest	5000	4506	494	2237237	123187	0.0353	0.9012	0.0679	0.0392	90.12	2113.79
CrossEval_TrainV2_TestV3_Full	Isolation Forest	10000	6603	3397	2234334	121090	0.0517	0.6603	0.0959	0.0783	66.03	4227.57
CrossEval_TrainV2_TestV3_Full	Isolation Forest	20000	11429	8571	2229160	116264	0.0895	0.5714	0.1548	0.1566	57.14	8455.14
CrossEval_TrainV2_TestV3_Full	Isolation Forest	30000	20374	9626	2228105	107319	0.1596	0.6791	0.2584	0.2349	67.91	12682.72

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RESULT: TRAIN V2 -> TEST V3

Observation:

- Supervised models generalize well
- Precision@K remains high.

Explanation:

- Patterns learned from V2 effectively cover the variations found in V3.

TRAIN V3 → TEST V2

Scenario	Model	K	TP	FP	TN	FN	Recall@K	Precision@K	F1_Score	Max_Possible_Recall	Efficiency (%)	Alerts/1M_flows
CrossEval_TrainV3_TestV2_Full	Random Forest	50	38	12	2295210	95015	0.0004	0.76	0.0008	0.0005	76	20.92
CrossEval_TrainV3_TestV2_Full	Random Forest	100	88	12	2295210	94965	0.0009	0.88	0.0018	0.0011	88	41.84
CrossEval_TrainV3_TestV2_Full	Random Forest	500	425	75	2295147	94628	0.0045	0.85	0.0089	0.0053	85	209.18
CrossEval_TrainV3_TestV2_Full	Random Forest	1000	425	575	2294647	94628	0.0045	0.425	0.0088	0.0105	42.5	418.36
CrossEval_TrainV3_TestV2_Full	Random Forest	5000	2791	2209	2293013	92262	0.0294	0.5582	0.0558	0.0526	55.82	2091.81
CrossEval_TrainV3_TestV2_Full	Random Forest	10000	7791	2209	2293013	87262	0.082	0.7791	0.1483	0.1052	77.91	4183.62
CrossEval_TrainV3_TestV2_Full	Random Forest	20000	16879	3121	2292101	78174	0.1776	0.844	0.2934	0.2104	84.4	8367.24
CrossEval_TrainV3_TestV2_Full	Random Forest	30000	26620	3380	2291842	68433	0.2801	0.8873	0.4257	0.3156	88.73	12550.86
CrossEval_TrainV3_TestV2_Full	XGBoost	50	24	26	2295196	95029	0.0003	0.48	0.0005	0.0005	48	20.92
CrossEval_TrainV3_TestV2_Full	XGBoost	100	74	26	2295196	94979	0.0008	0.74	0.0016	0.0011	74	41.84
CrossEval_TrainV3_TestV2_Full	XGBoost	500	185	315	2294907	94868	0.0019	0.37	0.0039	0.0053	37	209.18
CrossEval_TrainV3_TestV2_Full	XGBoost	1000	188	812	2294410	94865	0.002	0.188	0.0039	0.0105	18.8	418.36
CrossEval_TrainV3_TestV2_Full	XGBoost	5000	2881	2119	2293103	92172	0.0303	0.5762	0.0576	0.0526	57.62	2091.81
CrossEval_TrainV3_TestV2_Full	XGBoost	10000	7872	2128	2293094	87181	0.0828	0.7872	0.1499	0.1052	78.72	4183.62
CrossEval_TrainV3_TestV2_Full	XGBoost	20000	17236	2764	2292458	77817	0.1813	0.8618	0.2996	0.2104	86.18	8367.24
CrossEval_TrainV3_TestV2_Full	XGBoost	30000	26534	3466	2291756	68519	0.2791	0.8845	0.4244	0.3156	88.45	12550.86
CrossEval_TrainV3_TestV2_Full	Isolation Forest	50	45	5	2295217	95008	0.0005	0.9	0.0009	0.0005	90	20.92
CrossEval_TrainV3_TestV2_Full	Isolation Forest	100	90	10	2295212	94963	0.0009	0.9	0.0019	0.0011	90	41.84
CrossEval_TrainV3_TestV2_Full	Isolation Forest	500	332	168	2295054	94721	0.0035	0.664	0.0069	0.0053	66.4	209.18
CrossEval_TrainV3_TestV2_Full	Isolation Forest	1000	649	351	2294871	94404	0.0068	0.649	0.0135	0.0105	64.9	418.36
CrossEval_TrainV3_TestV2_Full	Isolation Forest	5000	3297	1703	2293519	91756	0.0347	0.6594	0.0659	0.0526	65.94	2091.81
CrossEval_TrainV3_TestV2_Full	Isolation Forest	10000	6178	3822	2291400	88875	0.065	0.6178	0.1176	0.1052	61.78	4183.62
CrossEval_TrainV3_TestV2_Full	Isolation Forest	20000	8647	11353	2283869	86406	0.091	0.4324	0.1503	0.2104	43.24	8367.24
CrossEval_TrainV3_TestV2_Full	Isolation Forest	30000	8871	21129	2274093	86182	0.0933	0.2957	0.1419	0.3156	29.57	12550.86

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RESULT: TRAIN V3 -> TEST V2

Observation:

- Significant performance drop.

Explanation:

- Caused by Feature/Scale mismatch and attack distribution shifts highlighted in the Domain Shift analysis.

LINK: DOMAIN SHIFT & CROSS-DATASET

- Correlation: The drop in cross-dataset performance directly aligns with the 1.0 Domain Classifier score.
- Conclusion: Performance degradation is a result of data shift, not a fundamental model failure.

REAL-WORLD IMPLICATION

Cannot rely only on accuracy

Need:

- robust features
- proper evaluation
- domain awareness

LIMITATIONS & FUTURE DIRECTIONS

Limitations:

- Cross-dataset evaluation shows strong distribution mismatch
- No real-time deployment validation
- Supervised models struggle with unseen / unknown attacks
- Deep learning comparison left for future work

Future Directions:

- Cross-environment evaluation
- Low-latency IDS deployment
- Domain adaptation techniques

CONCLUSION

Project Summary

- IF serves as an anomaly layer; RF/XGB are the primary practical models.
- Feature ablation reduces shortcuts and increases model robustness.
- Domain Shift is a critical factor identified for real-world deployment.

Possible Extensions

- Evaluate the pipeline on additional network environments.
- Further analyze model behavior under domain shift.
- Explore deployment-oriented threshold selection.
- Investigate more robust cross-domain settings.

“A good IDS model is not just accurate — it must be robust, interpretable, and deployable in changing environments.”

FPT Education

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THANK YOU